PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

CONVERSATIONAL IMAGE RECOGNITION CHATBOT

A PROJECT REPORT

Submitted by

K KEERTHI – 20221ISE0046

ADITI N – 20221ISE0085

RAVIRAJ P MATH – 20221ISE0055

Under the guidance of,

Mr. JAI KUMAR B

BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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BONAFIDE CERTIFICATE

Certified that this report "**CONVERSATIONAL IMAGE RECOGNITION CHATBOT**" Is a Bonafide work of "K KEERTHI (20221ISE0046), ADITI N(20221ISE0085), RAVIRAJ P MATH (20221ISE0055)," who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in **INFORMATION SCIENCE AND ENGINEERING** during 2025-26.

2/12/25

02/12/25

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PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We the students of final year B. Tech in INFORMATION SCIENCE ENGINEERING at Presidency University, Bengaluru, named K KEERTHI, ADITI N, RAVIRAJ P MATH, hereby declare that the project work titled **"CONVERSATIONAL IMAGE RECOGNITION CHATBOT"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B. Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

K KEERTHI	USN: 20221ISE0046	Keerthi
ADITI N	USN: 20221ISE0085	Aditi
RAVIRAJ P MATH	USN: 20221ISE0055	

PLACE: BENGALURU

DATE: 02/12/2025

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K KEERTHI

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Abstract

Artificial intelligence has changed how we relate with technology, and conversational AI is one of them as it assists individuals with chores, starting with personal assistance and customer care. However, classical chatbots have drawbacks: they are good in text comprehension and text generation, but they cannot understand visual information, including pictures, schemes, or any sophisticated scenes. This makes them very little useful in most of the real-life scenarios where rationalization is composed of seeing as much as of understanding. To combat this, the project proposes a multimodal chatbot which has the capability of viewing as well as chatting. It incorporates object detection, visual question answering, context-sensitive dialogue as well as image captioning into one system. And the output systems can process a view, answer questions about that view, and carry out extended turn-taking dialogs based on understanding a view, descriptive caption languages with BLIP/BLIP-2, and object detection and labelling with YOLOv8. It is designed to be reconfigurable with each part being upgradeable or enhanced, and this is because of the modular design that is future-proof in the event of advancements in AI. Long-term experimentation revealed that the chatbot is able to identify various objects and make purposeful captions with precision when amending multiple rounds of interaction and conducting contextual relevant reactions. It is therefore applicable in e-commerce product identification, industry monitoring, accessibility tools to users with visual impairments and in educational interactive solutions. To summarize, this project exhibits a multimodal real-time chatbot that is a combination of conversational intelligence and visual perception. It combines the vision and the language, and therefore, it opens the way to more human, interactive, and intuitive communication between humans and machines.